

Business Insights Report

Epic 2 E-commerce Dataset - Statistical Analysis & Visualization

Generated 2026-09-27 | Epic 3 checkpoint (Days 11-14 combined)

Dataset: 150 orders, 3 numeric columns analyzed.

Executive Summary - Descriptive Statistics

Column	Mean	Median	Mode	Std (sample)	Range	IQR
quantity	5.20	6.00	6.00	2.58	8.00	4.75
unit_price	270.20	284.65	284.65	136.85	491.87	226.97
discount_pct	9.63	10.00	10.00	3.10	10.00	4.50

Dataset: 150 orders (Epic 2 cleaned + merged dataset). All values computed from scratch (Day 11) and cross-validated against NumPy/SciPy/pandas.

Epic 2 Dataset - Statistical Dashboard

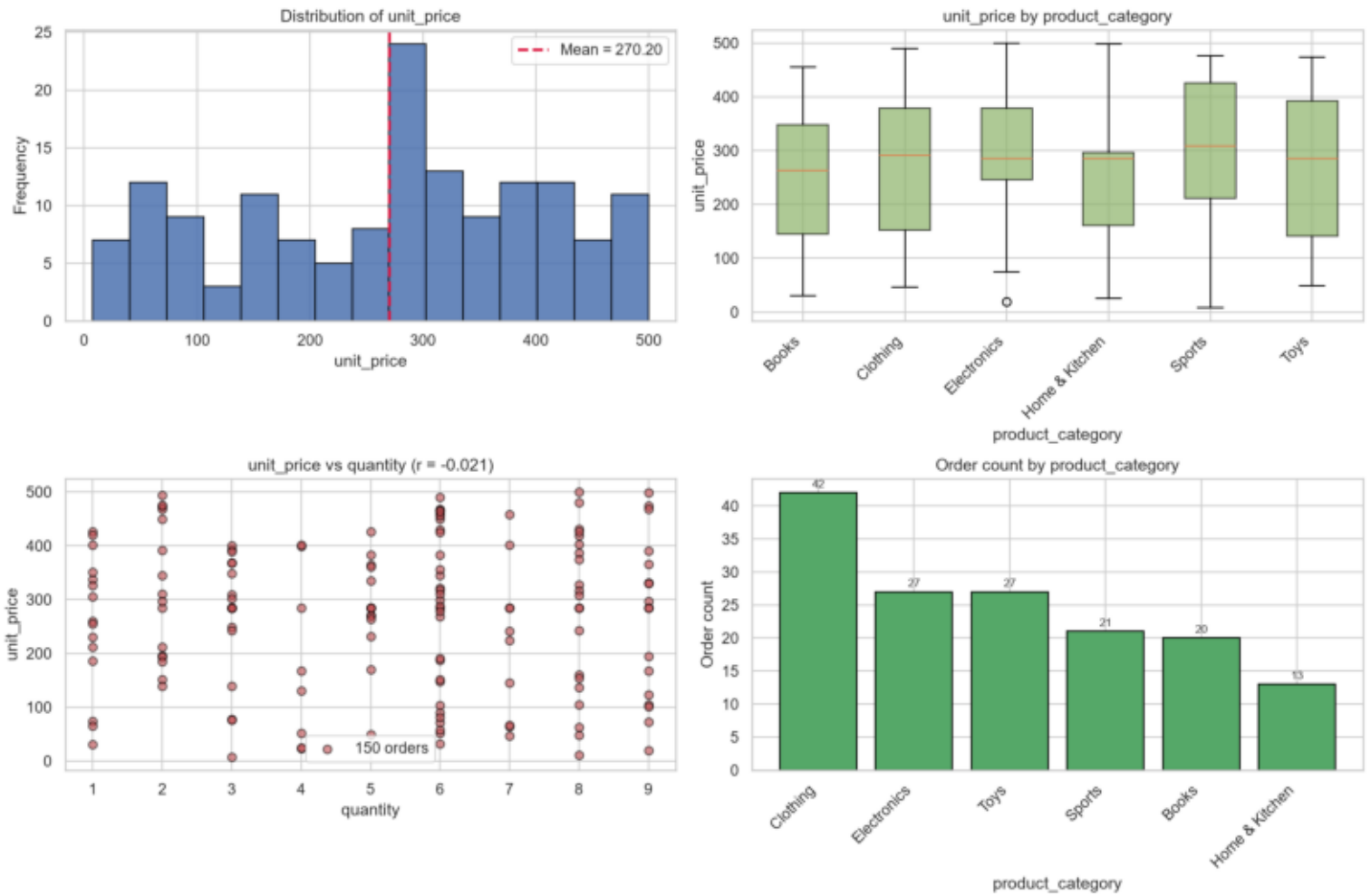


Figure 1. Statistical dashboard overview (histogram, boxplot, scatter, bar chart).

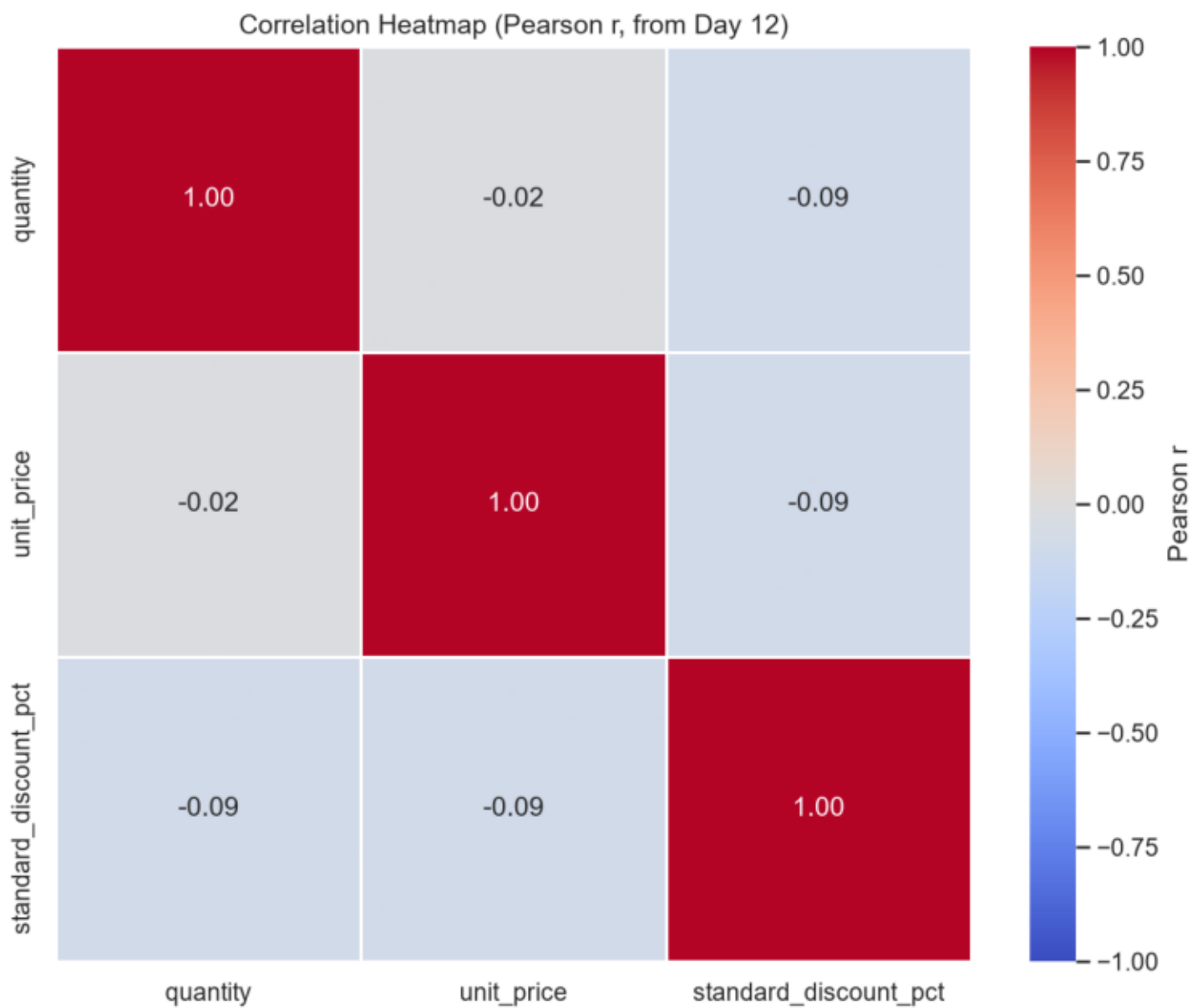


Figure 2. Pearson correlation heatmap across numeric columns.

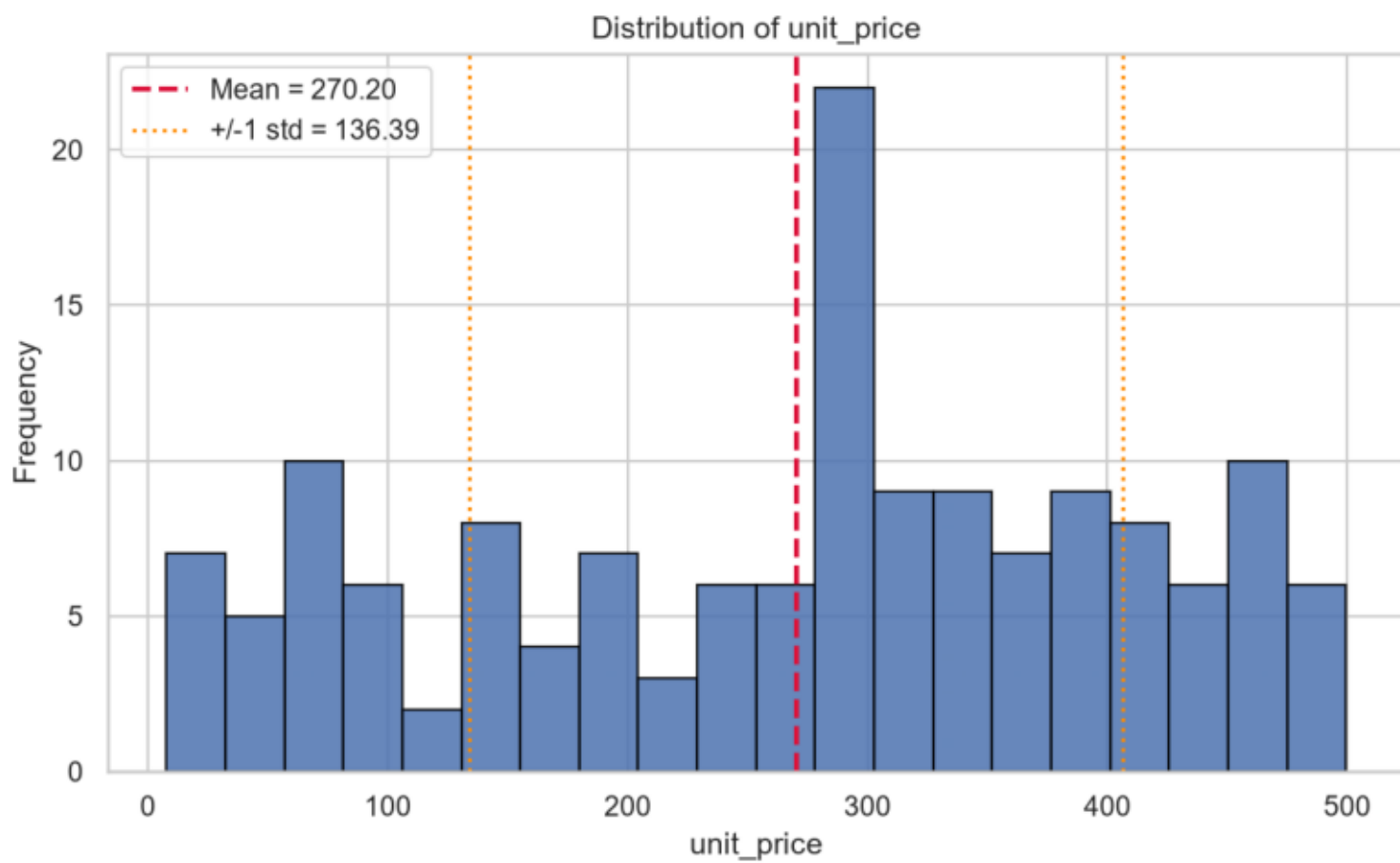


Figure 3. Distribution of unit_price.

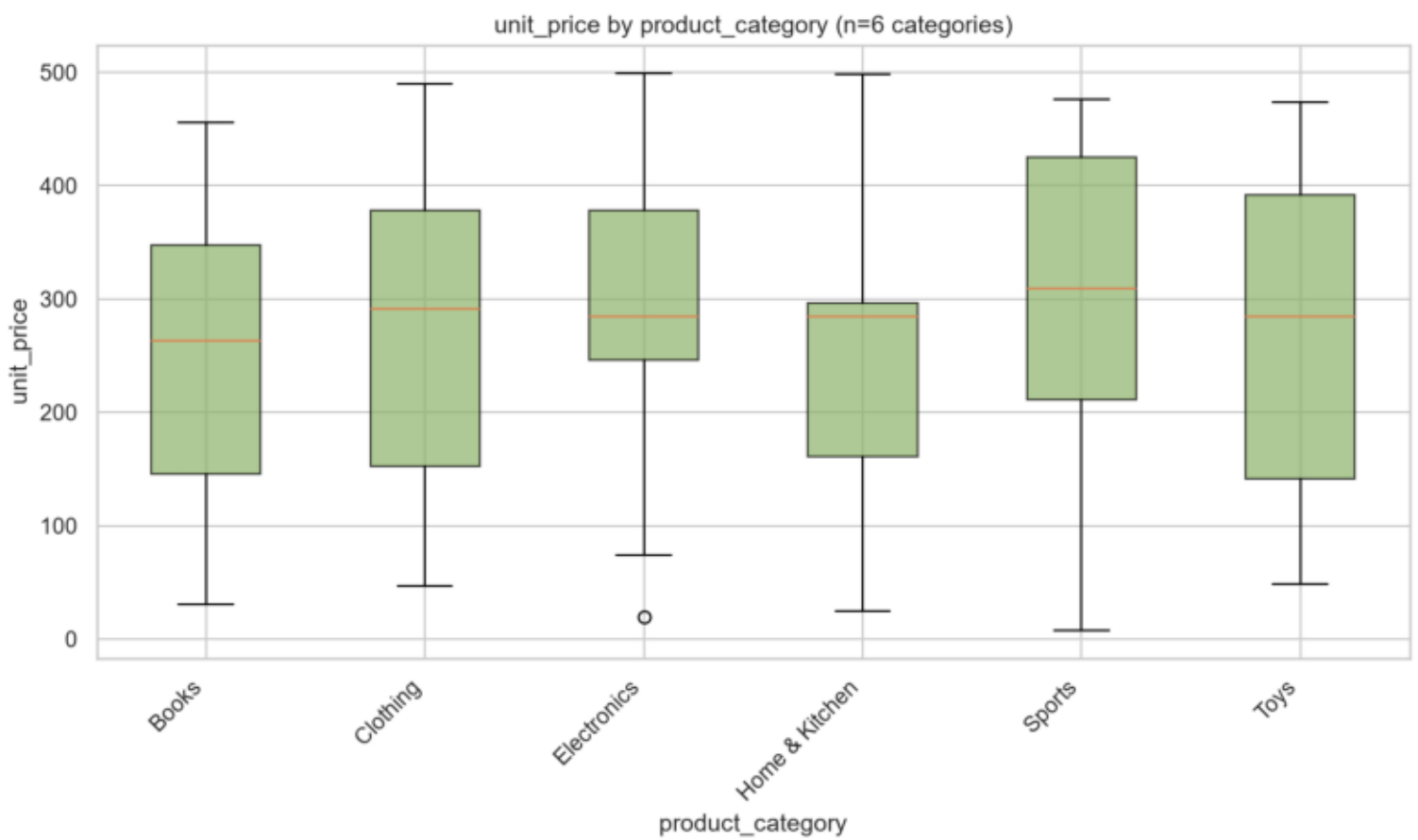


Figure 4. unit_price spread by product_category.

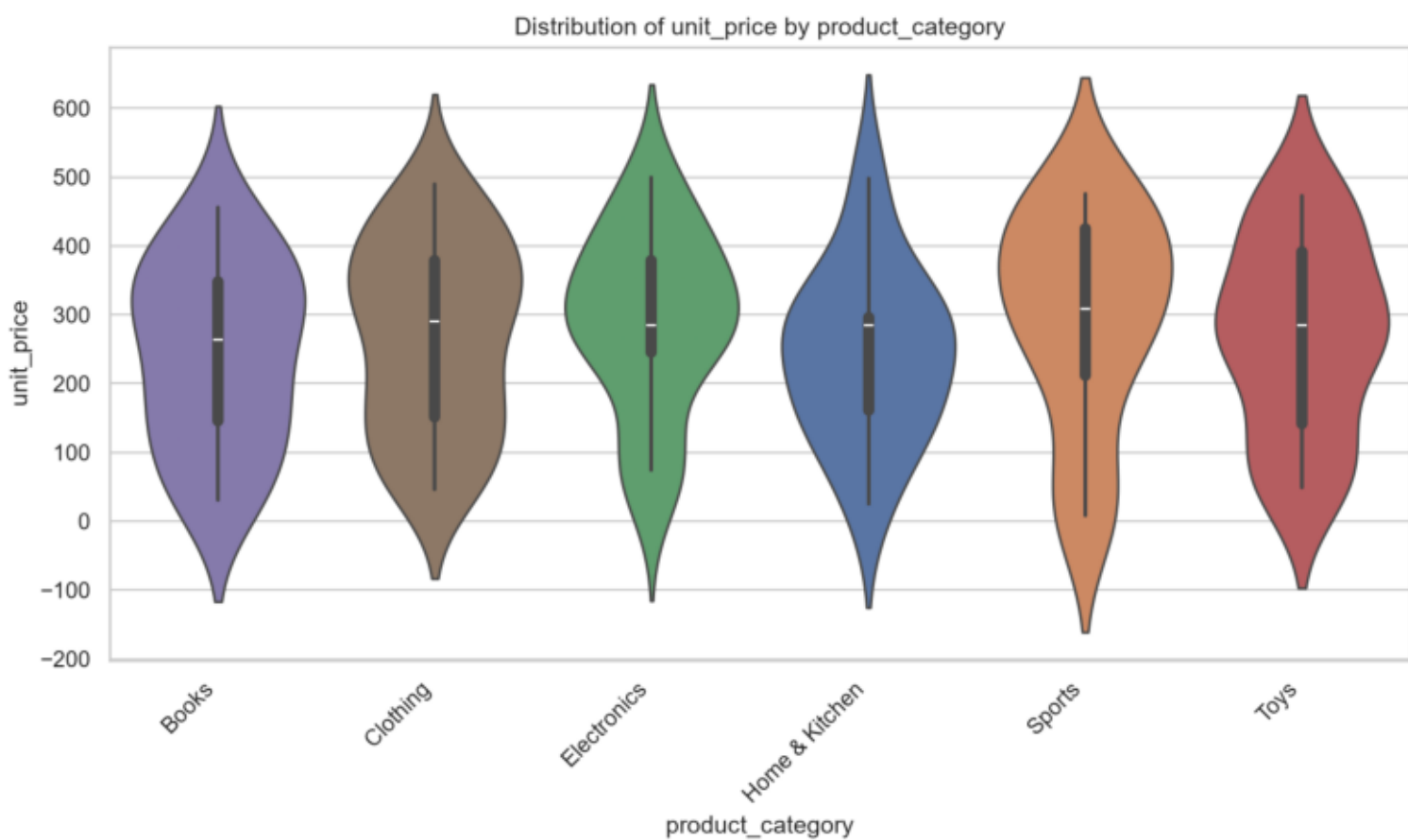


Figure 5. unit_price density by product_category (Seaborn violin).

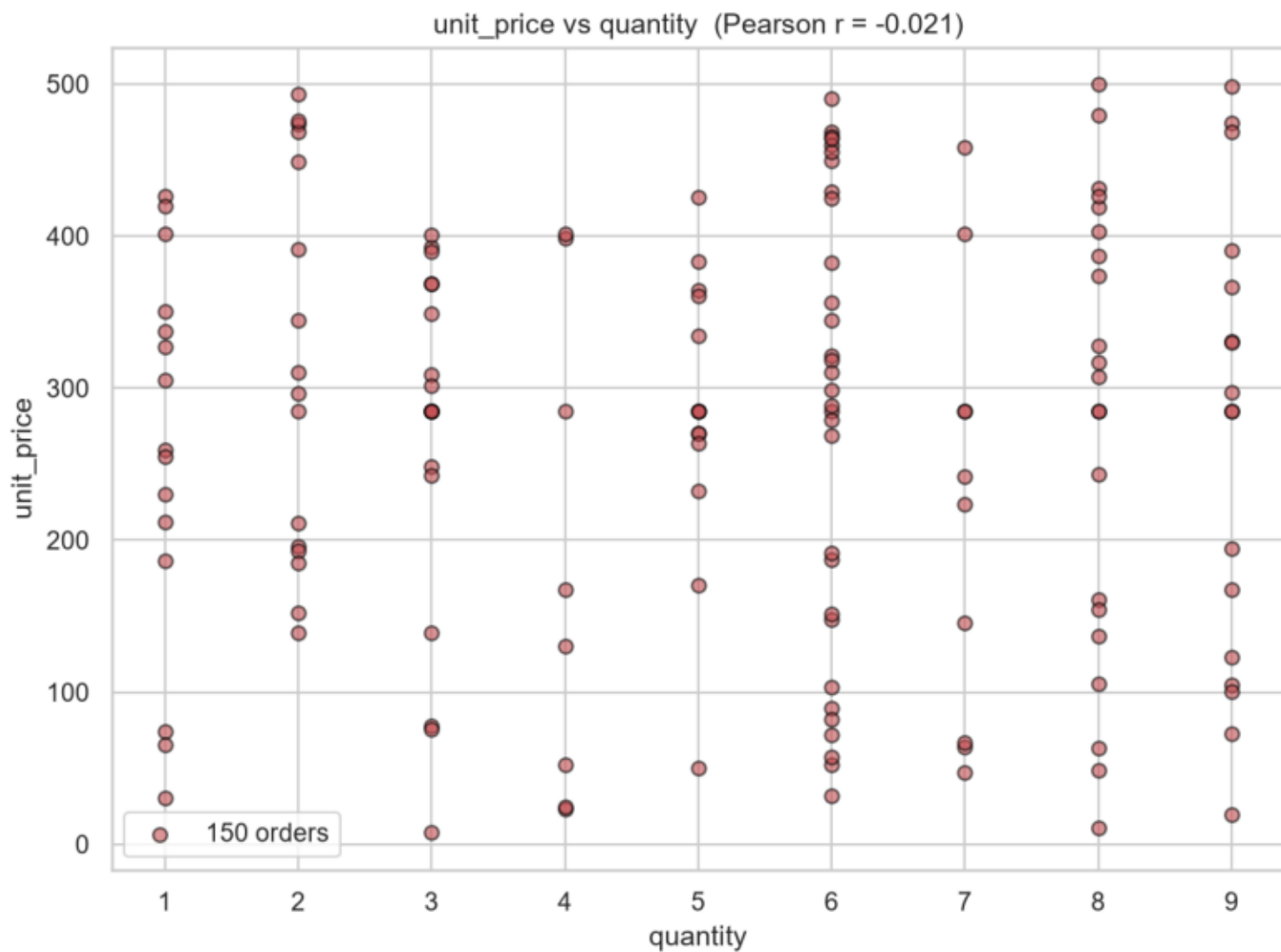


Figure 6. quantity vs. unit_price scatter.

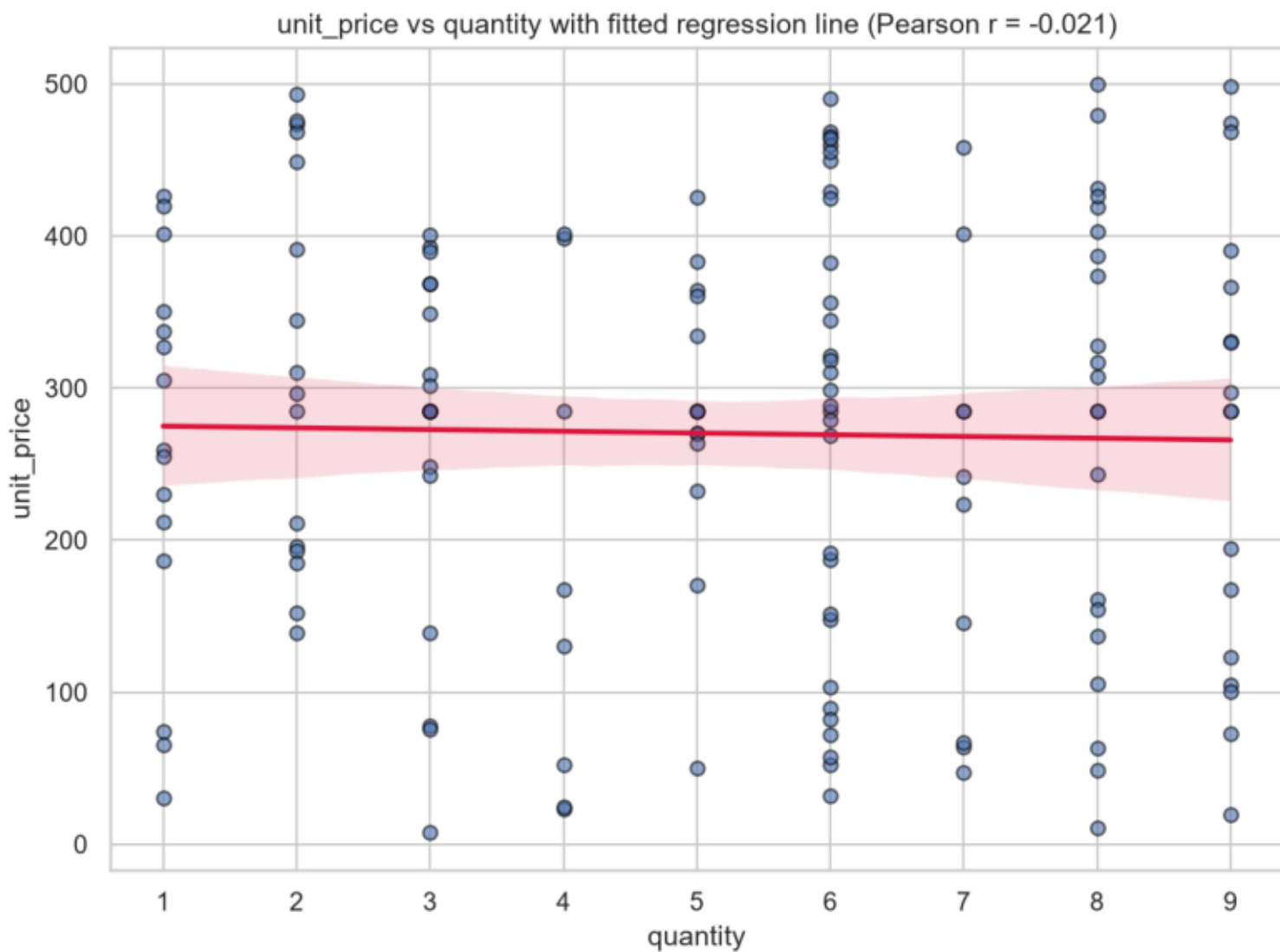


Figure 7. quantity vs. unit_price with fitted regression line.

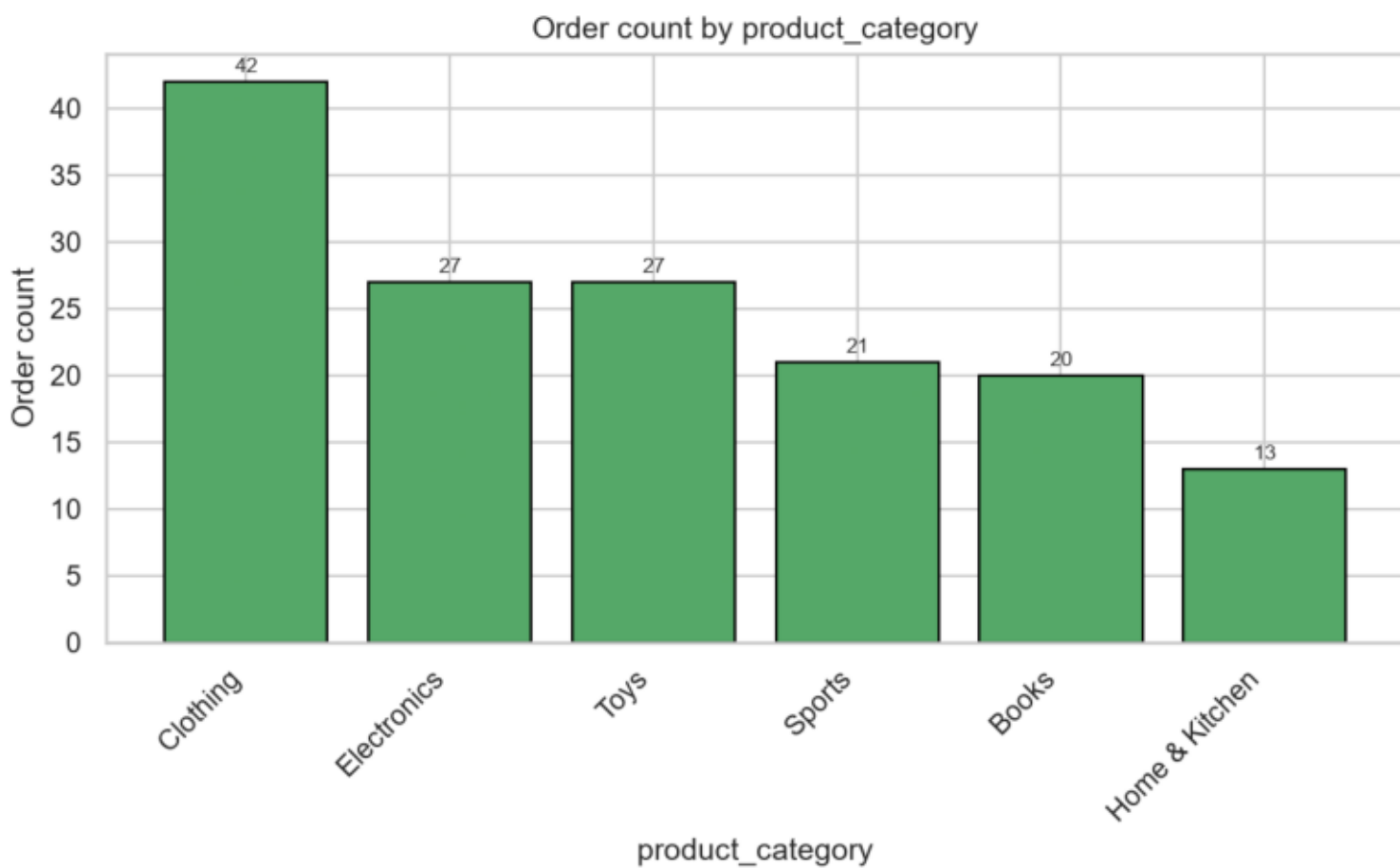


Figure 8. Order count by product_category.

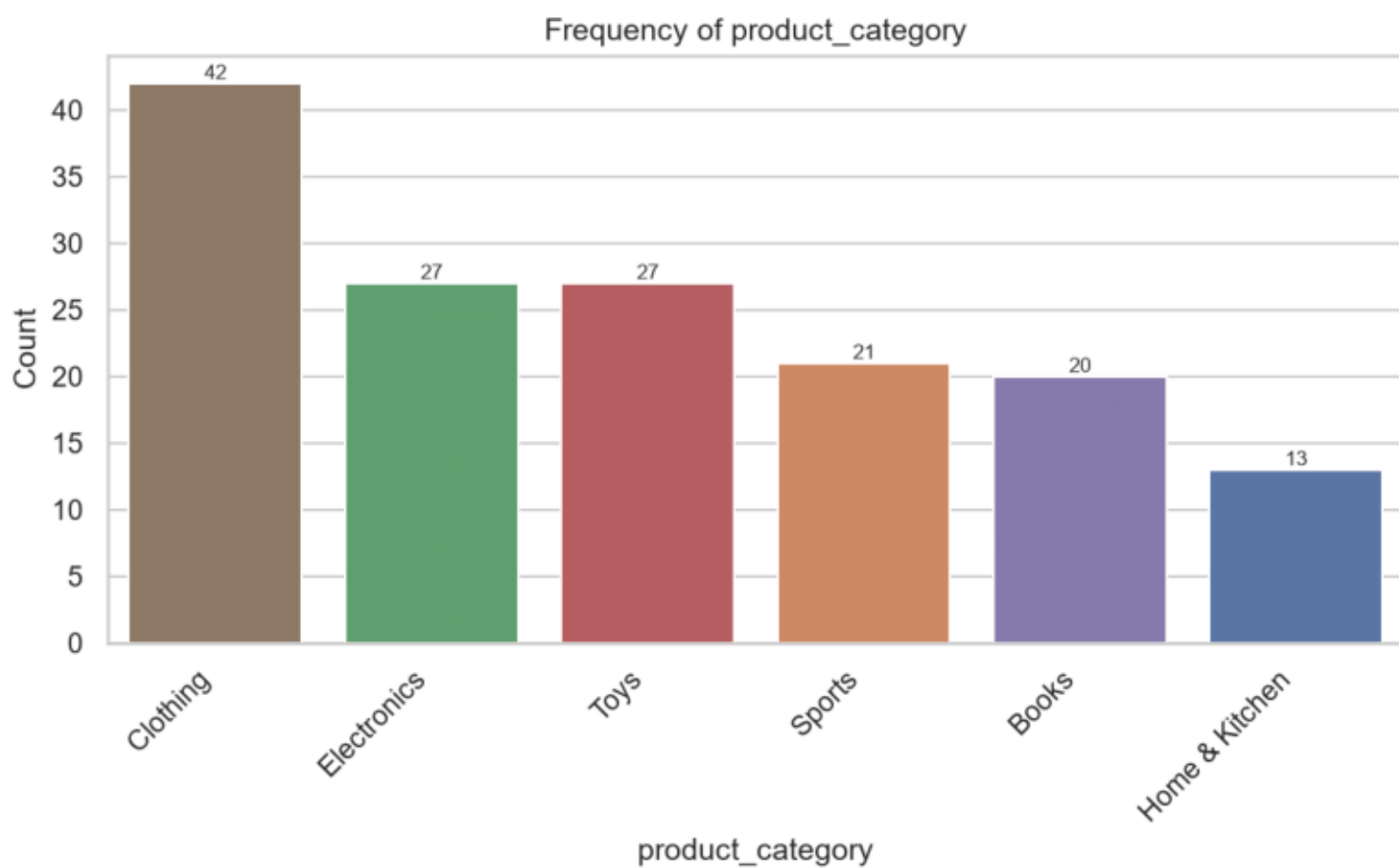


Figure 9. Category frequency (Seaborn countplot).

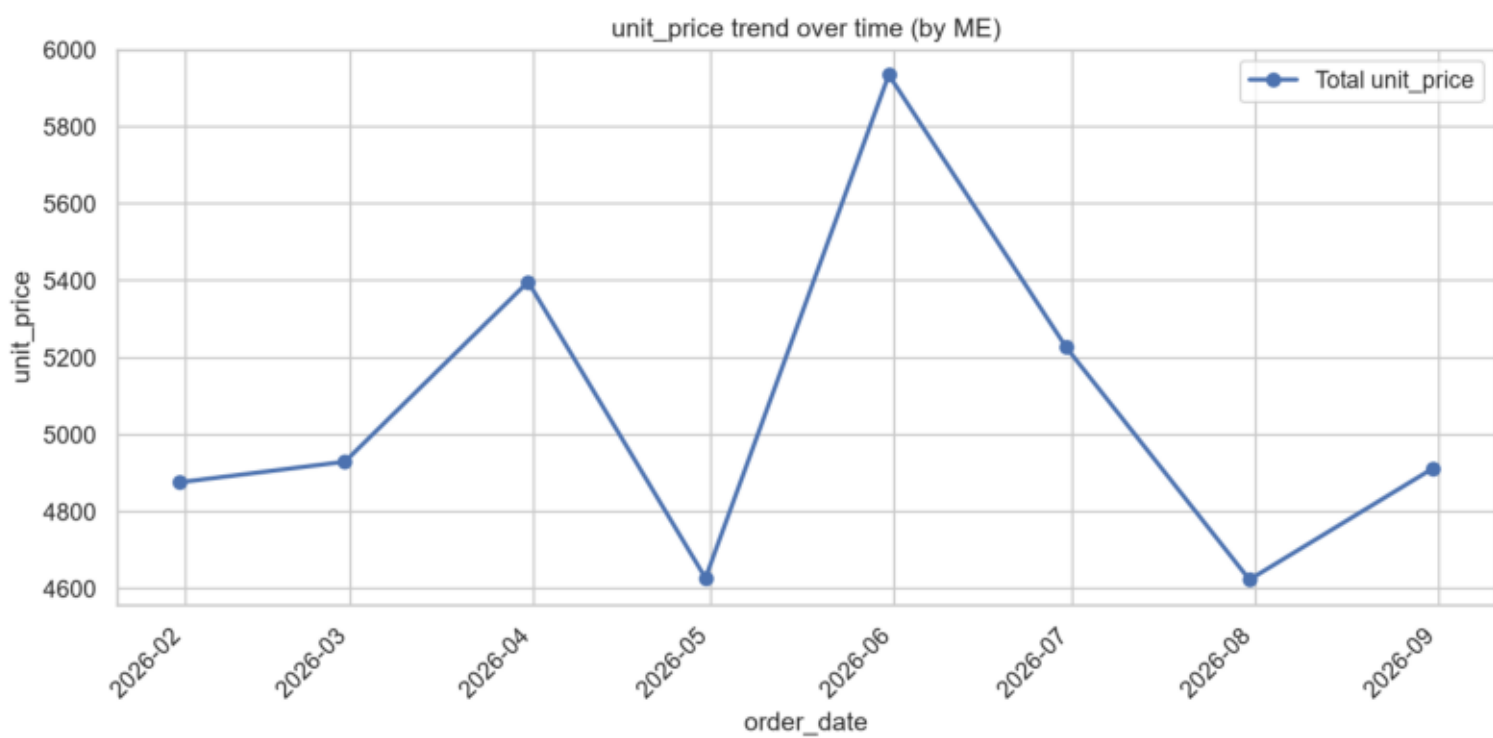


Figure 10. Monthly unit_price total trend.

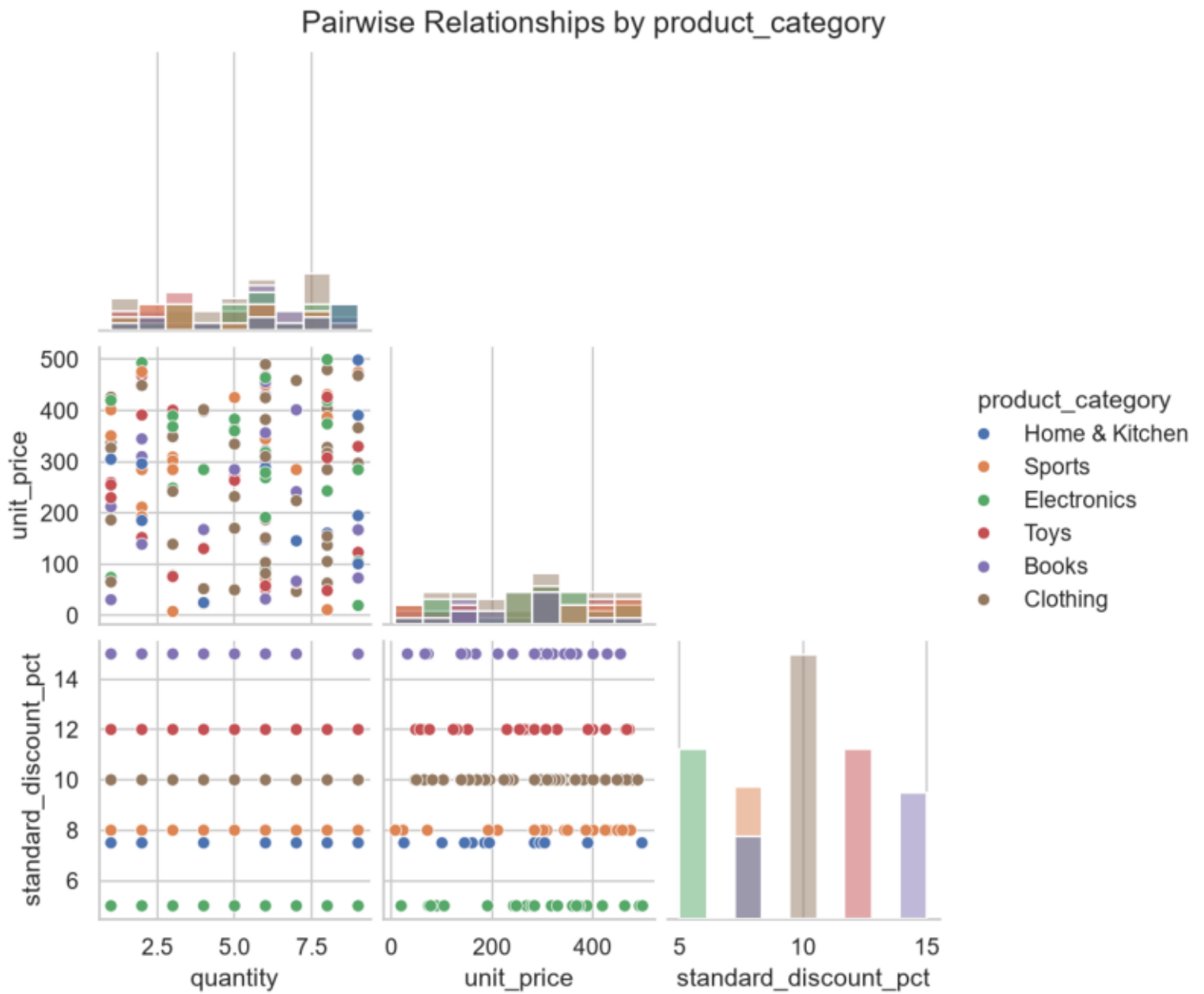


Figure 11. Pairwise numeric relationships, colored by product_category.

Key Findings

1. Orders average \$270.20 per unit (median \$284.65), with a standard deviation of \$136.85 - prices are widely spread across a range of \$491.87, not clustered around one typical value.
2. Typical order quantity is 5.2 units (mode 6), with an IQR of 4.75 units - most orders fall within a fairly narrow band.
3. The strongest pairwise relationship found is between quantity and standard_discount_pct (Pearson $r = -0.092$), and even that is weak. The weakest is between quantity and unit_price ($r = -0.021$). No numeric column pair shows a meaningful linear relationship in this dataset.
4. Shapiro-Wilk testing rejects normality ($p < 0.05$) for quantity and unit_price - downstream analysis should not assume a bell-curve shape for these columns.
5. Z-score outlier screening ($\pm 2\sigma$) flagged 0 extreme value(s) across quantity, unit_price, standard_discount_pct - pricing and quantities are consistent, with no runaway values distorting the averages.
6. Clothing is the highest-volume category with 42 orders, versus only 13 for Home & Kitchen - a 3.2x difference in order volume between the busiest and quietest category.
7. Sports has the highest median unit_price (\$309.12) among categories, while Books has the lowest (\$263.10) - a \$46.02 gap that the boxplot/violin charts make visible at a glance.

Recommendations

1. Because no numeric column pair shows a meaningful linear (or monotonic) relationship, do not rely on quantity, price, or discount alone to predict one another - collect additional features (customer segment, marketing channel, seasonality) before attempting predictive modeling on this data.
2. Since quantity and unit_price both fail normality testing, avoid statistical methods that assume a normal distribution (e.g. raw z-tests or Pearson-based significance tests without checking assumptions) - prefer non-parametric methods (Spearman correlation, rank-based tests) or transform the data first.
3. Investigate why Home & Kitchen has such a low order volume relative to Clothing (13 vs. 42 orders) - determine whether this reflects genuine low demand or a fixable gap in marketing/inventory support before deprioritizing the category.